An exact projective-weight criterion for topological ergodicity of coechelon shifts

Candidate full answer to Open Problem (1) of arXiv:1911.07513

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Status. Candidate full solution, likely valid, of Open Problem (1). The theorem below gives a necessary and sufficient condition directly in the defining weights. It does not claim to solve the source's separate Open Problem (2).

1 Source question

The source is J. Bonet, T. Kalmes, and A. Peris, *Dynamics of shift operators on non-metrizable sequence spaces*, arXiv:1911.07513v2 [1]. It appeared in *Revista Matemática Iberoamericana* 37(6) (2021), 2373–2397. Its final subsection asks for a condition that completely characterizes topological ergodicity of the backward shift on a Köthe coechelon space. Figure 1 is a crop of page 20 of the source PDF.

4.5. **Open problems.** In this final subsection we mention the following natural questions which arise from our results.
(1) In Proposition 2.9 we gave a sufficient condition for topological ergodicity. However, this condition is not necessary by Proposition 4.5. Which condition completely characterizes topological ergodicity of B?
(2) Is there a nuclear Köthe coechelon space $k_p(V)$ on which the backward shift B is topologically ergodic but not sequentially hypercyclic? Such an example would be a strengthening of both Propositions 4.3 and 4.4.

Figure 1: The two open problems stated in the source. This packet answers Problem (1).

2 Setting

Let $V = (v^{(m)})_{m\in\mathbb{N}}$ be a decreasing sequence of strictly positive weights on $\mathbb{N}$, and let $p = 0$ or $1 \le p < \infty$. Assume that the backward shift

$$Be_1 = 0, \qquad Be_j = e_{j-1} \quad (j \ge 2)$$

is continuous on the Köthe coechelon space $k_p(V)$. As in the source, its projective family is

$$\overline{V} = \left\{ \bar{v} \in [0,\infty)^{\mathbb{N}} : \text{for every } m \text{ there is } a_m > 0 \text{ such that } \bar{v}_j \le a_m v_j^{(m)} \text{ for all } j \right\}. \tag{2.1}$$

The topology of $k_p(V) = K_p(\overline{V})$ is generated by

$$q_{\bar{v}}(x) = \begin{cases} \left(\sum_j |x_j|^p \bar{v}_j^p\right)^{1/p}, & 1 \le p < \infty, \\ \sup_j |x_j| \bar{v}_j, & p = 0, \end{cases} \qquad \bar{v} \in \overline{V}. \tag{2.2}$$

A set $A \subseteq \mathbb{N}$ is *syndetic* if its gaps are bounded. The operator B is topologically ergodic when

$$N_B(U,W) = \{n \geq 0 : B^n(U) \cap W \neq \varnothing\} \tag{2.3}$$

is syndetic for every pair of nonempty open sets U, W.

3 Proof intuition

For necessity, test ergodicity with a small neighborhood of zero and a small neighborhood of e_1. A return at time n forces the $(n+1)$st coordinate of the starting vector to be close to one. Smallness in the projective seminorm then forces the projective weight at $n+1$ to be small. Thus the small-coordinate set of every projective weight must be syndetic.

For sufficiency, start with finitely supported points in two arbitrary open sets. Only finitely many inverse images of one zero neighborhood are needed to implant the second point far out in the first one. A finite intersection of projective seminorm balls is dominated by one projective seminorm ball. The assumed syndeticity therefore supplies syndetically many locations at which the implantation works.

Finally, all projective weights can be tested at once through the canonical weights

$$\bar{v}_j^a = \inf_m a_m v_j^{(m)}. \tag{3.1}$$

This converts the topological condition into a formula involving only the given matrix V.

4 Exact characterization

Theorem 1 (answer to Open Problem (1))**.** *Under the assumptions above, the following are equivalent.*

(i) The backward shift B is topologically ergodic on $k_p(V)$.

(ii) For every $\bar{v} \in \overline{V}$ and every $\varepsilon > 0$, the set

$$S(\bar{v}, \varepsilon) = \{j \in \mathbb{N} : \bar{v}_j < \varepsilon\} \tag{4.1}$$

is syndetic.

(iii) For every sequence $a = (a_m)_{m\in\mathbb{N}}$ of positive scalars and every $\varepsilon > 0$, the set

$$\left\{ j \in \mathbb{N} : \inf_{m\in\mathbb{N}} a_m v_j^{(m)} < \varepsilon \right\} \tag{4.2}$$

is syndetic.

In particular, the criterion is independent of p.

Proof. (i)⇒(ii). Fix $\bar{v} \in \overline{V}$ and $\varepsilon > 0$. Choose $c > 0$ and put

$$\widetilde{v} = \bar{v} + c\,\mathbf{1}_{\{1\}}.$$

Then $\widetilde{v} \in \overline{V}$, because the added sequence has finite support, and $\widetilde{v}_1 > 0$. Consider the nonempty open sets

$$U = \{x : q_{\widetilde{v}}(x) < \varepsilon/2\}, \qquad W = e_1 + \{z : q_{\widetilde{v}}(z) < \widetilde{v}_1/2\}. \tag{4.3}$$

If $n \in N_B(U,W)$, take $x \in U$ with $B^n x \in W$. Since $(B^n x)_1 = x_{n+1}$, the second condition in (4.3) gives

$$|x_{n+1} - 1|\widetilde{v}_1 < \widetilde{v}_1/2, \quad \text{hence} \quad |x_{n+1}| > 1/2.$$

The first condition gives

$$|x_{n+1}|\widetilde{v}_{n+1} \le q_{\widetilde{v}}(x) < \varepsilon/2.$$

Consequently $\bar{v}_{n+1} \le \widetilde{v}_{n+1} < \varepsilon$. We have proved

$$N_B(U,W) + 1 \subseteq S(\bar{v}, \varepsilon). \tag{4.4}$$

The left side is syndetic by (i), and any superset of a syndetic set is syndetic. This proves (ii). Notice that the finite-coordinate correction also covers projective weights with $\bar{v}_1 = 0$.

(ii)⇒(i). Let U, W be nonempty open subsets of $k_p(V)$. The finitely supported sequences are dense, so there are finitely supported $x \in U$, $y \in W$ and an absolutely convex zero neighborhood G such that

$$x + G \subseteq U, \qquad y + G \subseteq W. \tag{4.5}$$

Choose $s \ge 1$ with both supports contained in $\{1, \dots, s\}$, write $y = \sum_{k=1}^{s} y_k e_k$, and set

$$M = 1 + \sum_{k=1}^{s} |y_k|, \qquad G_1 = \bigcap_{r=0}^{s-1} B^{-r}(G/M). \tag{4.6}$$

Continuity of B makes G_1 a zero neighborhood. By the projective description, there are an integer L, weights $\bar{v}^{(1)}, \dots, \bar{v}^{(L)} \in \overline{V}$, and positive numbers $\delta_1, \dots, \delta_L$ such that

$$\bigcap_{\ell=1}^{L} \{z : q_{\bar{v}^{(\ell)}}(z) < \delta_\ell\} \subseteq G_1. \tag{4.7}$$

Define a single projective weight by

$$\omega_j = \begin{cases} \left(\displaystyle\sum_{\ell=1}^{L} (\bar{v}_j^{(\ell)}/\delta_\ell)^p \right)^{1/p}, & 1 \le p < \infty, \\ \displaystyle\max_{1 \le \ell \le L} \bar{v}_j^{(\ell)}/\delta_\ell, & p = 0. \end{cases} \tag{4.8}$$

Finite combination in (2.1) shows that $\omega \in \overline{V}$. Moreover, $\omega_j < 1$ implies $q_{\bar{v}^{(\ell)}}(e_j) < \delta_\ell$ for every ℓ, and hence $e_j \in G_1$ by (4.7). Thus (ii) shows that

$$J = \{j : \omega_j < 1\} \tag{4.9}$$

is syndetic and $e_j \in G_1$ for every $j \in J$.

For $j \in J$ with $j \ge 2s$, put

$$h_j = \sum_{k=1}^{s} y_k e_{j-s+k} = \sum_{k=1}^{s} y_k B^{s-k} e_j. \tag{4.10}$$

By (4.6), every $B^{s-k}e_j$ belongs to G/M. Absolute convexity and $\sum_k |y_k| < M$ therefore give $h_j \in G$. With $n = j - s \ge s$,

$$B^n(x + h_j) = B^n h_j = y. \tag{4.11}$$

Equations (4.5) and (4.11) show that

$$\{j - s : j \in J,\ j \geq 2s\} \subseteq N_B(U, W).$$

Deleting a finite prefix and translating preserve syndeticity. Hence the right side is syndetic, proving (i).

(ii)⇔(iii). For every positive sequence a, the canonical weight in (3.1) belongs to $\overline{V}$, because for each fixed r,

$$\bar{v}^{\,a}_j \leq a_r v^{(r)}_j \qquad (j \in \mathbb{N}).$$

Thus (ii) implies (iii). Conversely, if $\bar{v} \in \overline{V}$, choose $a_m > 0$ from (2.1). Then

$$\bar{v}_j \leq \inf_m a_m v^{(m)}_j = \bar{v}^{\,a}_j. \tag{4.12}$$

The small-coordinate set in (4.2) is therefore contained in $S(\bar{v}, \varepsilon)$. Under (iii) the former is syndetic, so the latter is syndetic as well. This proves (ii). □

5 Consistency checks and the remaining problem

The theorem specializes correctly to the source's sufficient condition. If for one row m every set $\{j : v^{(m)}_j < \eta\}$ is syndetic, then for $\bar{v} \leq a_m v^{(m)}$ that row's ε/a_m-small set is contained in $S(\bar{v}, \varepsilon)$. It also specializes correctly to mixing: the source characterizes mixing by $\bar{v}_j \to 0$ for every $\bar{v} \in \overline{V}$, in which case every set in (4.1) is cofinite. Unlike the fixed-row sufficient condition, Theorem 1 also accommodates the source's example where mixing holds but no single row has the required syndetic small sets.

The source's Open Problem (2) remains open in this packet, but the theorem gives an exact reduction. Combining (iii) with the source's sequential hypercyclicity criterion, one must construct a nuclear V such that

1. for every positive (a_m) and every $\varepsilon > 0$, the set in (4.2) is syndetic; while
2. for every m and every thick $I \subseteq \mathbb{N}$, the sequence $v^{(m)}_j$ fails to tend to zero as $j \to \infty$ through I.

This separates the still-missing construction from the now-complete ergodicity criterion.

6 Verification and novelty note

The proof was checked separately at the following points: projective weights may vanish; the finite sum/max in (4.8) stays in $\overline{V}$; the argument works for both $p = 0$ and finite p; and the shifts in (4.10)–(4.11) have the correct indices. Repository indexes and bounded exact-phrase, title/author, arXiv, and projective-weight searches performed through 13 August 2026 did not locate a later answer to Open Problem (1). This is a novelty check, not a proof of priority. Human review should focus first on the reduction of an arbitrary zero neighborhood to (4.7) and on the coordinate test (4.3).

References

[1] J. Bonet, T. Kalmes, and A. Peris, *Dynamics of shift operators on non-metrizable sequence spaces*, arXiv:1911.07513v2; Rev. Mat. Iberoam. 37(6) (2021), 2373–2397.